

SCAMPTON, United Kingdom

WMO: 033730

Lat: 53.3069N

Lon: 0.5481W

Elev: 202

StdP: 14.59

Time Zone: 0.00 (EUW)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.6	(c) 27.5	(d) 21.3	(e) 16.1	(f) 28.6	(g) 23.7	(h) 18.0	(i) 30.0	(j) 45.6	(k) 48.8	(l) 38.6	(m) 46.6	(n) 8.1	(o) 250	(p) 0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	78.3	64.9	74.7	63.6	71.8	62.1	66.8	74.4	65.1	72.1	63.5	69.9	10.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	89.7	69.7	62.3	84.5	67.8	60.8	80.3	66.4	31.4	74.7	30.1	72.4	28.9	69.8	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
33.8	27.3	23.4	18.7	85.1	5.2	3.2	14.9	87.4	11.9	89.2	9.0	91.0	5.2	93.3
			18.1	71.3	5.2	3.5	14.4	73.8	11.4	75.8	8.4	77.8	4.7	80.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	39.4	39.7	42.9	47.6	52.8	58.0	61.9	61.5	57.1	51.2	43.9	39.8
	DBStd	9.78	5.74	5.94	5.36	5.25	4.94	4.73	4.56	4.31	4.40	5.39	5.67	6.48
	HDD50	1546	329	291	227	107	27	2	0	0	2	49	192	320
	HDD65	5635	794	709	685	523	380	215	119	127	239	430	632	782
	CDD50	1440	1	2	7	34	112	243	370	355	216	86	11	3
	CDD65	53	0	0	0	0	0	7	24	18	3	1	0	0
	CDH74	371	0	0	0	2	6	62	182	97	18	4	0	0
	CDH80	51	0	0	0	0	0	8	22	19	2	0	0	0
(14) Wind	WSAvg	11.6	13.3	12.8	12.4	11.6	11.7	10.8	9.7	10.6	11.1	11.1	12.2	12.3
(15) Precipitation	PrecAvg	23.9	2.0	1.5	1.4	1.7	1.9	2.2	2.5	2.2	2.2	2.2	2.2	2.0
	PrecMax	32.3	4.5	2.8	2.8	5.4	4.3	8.2	4.9	5.5	4.7	4.0	3.8	3.8
	PrecMin	15.7	0.9	0.5	0.3	0.3	0.4	0.3	0.9	0.3	0.6	0.6	0.8	0.7
	PrecStd	4.6	0.9	0.6	0.7	1.3	1.0	1.7	1.2	1.2	1.1	0.8	0.8	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	57.0	61.2	69.9	73.9	80.7	83.0	83.0	76.5	69.0	60.1	56.2
		MCWB	52.1	52.6	52.9	57.1	62.5	67.7	67.0	67.2	63.6	61.5	57.1	53.1
	2%	DB	52.5	53.5	57.5	64.4	69.4	74.6	78.8	76.1	71.4	64.2	56.9	54.0
		MCWB	49.7	50.1	51.0	54.9	59.4	63.8	64.9	64.7	62.2	58.8	54.4	51.6
	5%	DB	50.2	50.9	54.7	60.5	65.7	70.6	75.5	72.8	68.0	61.2	54.7	51.8
		MCWB	47.8	48.0	49.6	52.5	56.9	61.7	63.9	62.7	60.3	57.4	52.4	49.5
	10%	DB	48.1	48.8	52.1	57.4	62.3	67.5	71.8	70.2	65.4	59.2	52.5	49.5
		MCWB	46.0	46.2	47.6	50.9	54.8	59.8	61.9	61.3	58.6	55.4	50.4	47.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.9	53.4	54.3	59.2	63.6	69.6	69.3	69.3	66.2	62.3	57.9	53.7
		MCDB	54.6	55.8	58.6	68.4	71.9	78.6	81.3	77.4	71.9	67.7	59.6	55.3
	2%	WB	50.3	50.5	52.1	56.1	60.6	65.6	66.5	66.2	63.5	60.1	54.8	51.9
		MCDB	52.2	53.0	56.2	62.6	67.9	72.0	74.7	73.3	69.3	63.2	56.5	53.8
	5%	WB	48.1	48.5	50.1	53.8	58.2	62.9	64.8	64.6	61.5	58.2	52.7	49.9
		MCDB	49.7	50.7	54.2	59.5	63.9	69.0	72.7	71.1	66.3	60.8	54.4	51.6
	10%	WB	46.2	46.5	48.2	51.5	56.1	60.8	63.2	62.9	59.9	56.1	50.7	47.7
		MCDB	48.0	48.5	51.7	56.7	60.9	66.3	70.1	68.4	64.1	58.6	52.3	49.5
(35) Mean Daily Temperature Range	MDBR		9.4	10.9	13.7	15.8	15.7	15.7	16.4	15.9	14.9	12.3	10.6	9.4
	5% DB	MCDBR	11.5	12.4	17.2	20.4	20.6	20.8	21.7	19.1	18.2	13.5	11.4	10.9
		MCWBR	10.7	9.9	11.8	12.2	12.1	12.0	10.3	9.4	10.2	9.3	10.1	10.0
	5% WB	MCDBR	11.4	12.1	16.2	19.7	18.6	19.1	19.2	17.0	16.1	12.7	11.2	10.9
		MCWBR	10.9	10.2	12.1	12.6	11.8	12.3	10.5	9.3	10.5	9.6	10.3	10.1
(40) Clear-Sky Solar Irradiance	taub		0.322	0.338	0.370	0.396	0.397	0.403	0.404	0.391	0.370	0.351	0.325	0.310
	taud		2.357	2.391	2.300	2.238	2.277	2.284	2.290	2.349	2.413	2.448	2.409	2.331
	Ebn at Noon		206	242	256	264	270	269	266	263	255	234	204	185
	Edh at Noon		19	24	32	39	39	40	39	35	29	23	17	16
(44) All-Sky Solar Radiation	RadAvg		216	432	773	1193	1494	1554	1511	1244	921	527	267	174
	RadStd		21	50	93	142	99	148	159	82	84	46	30	18

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page